

ASTA 607

COMPUTATIONAL STATISTICS 1

22425128

QUESTION 1

- a) The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.
- b) The crabs data is saved as a data frame made of 200 observations of 8 variables. The species (sp) variable is a factor with two levels, sex is also a factor with 2 levels. Index is an integer and the remaining are numeric.

```
Codes:
library(MASS)
library(psych)
data("crabs")
str(crabs)
```

c)

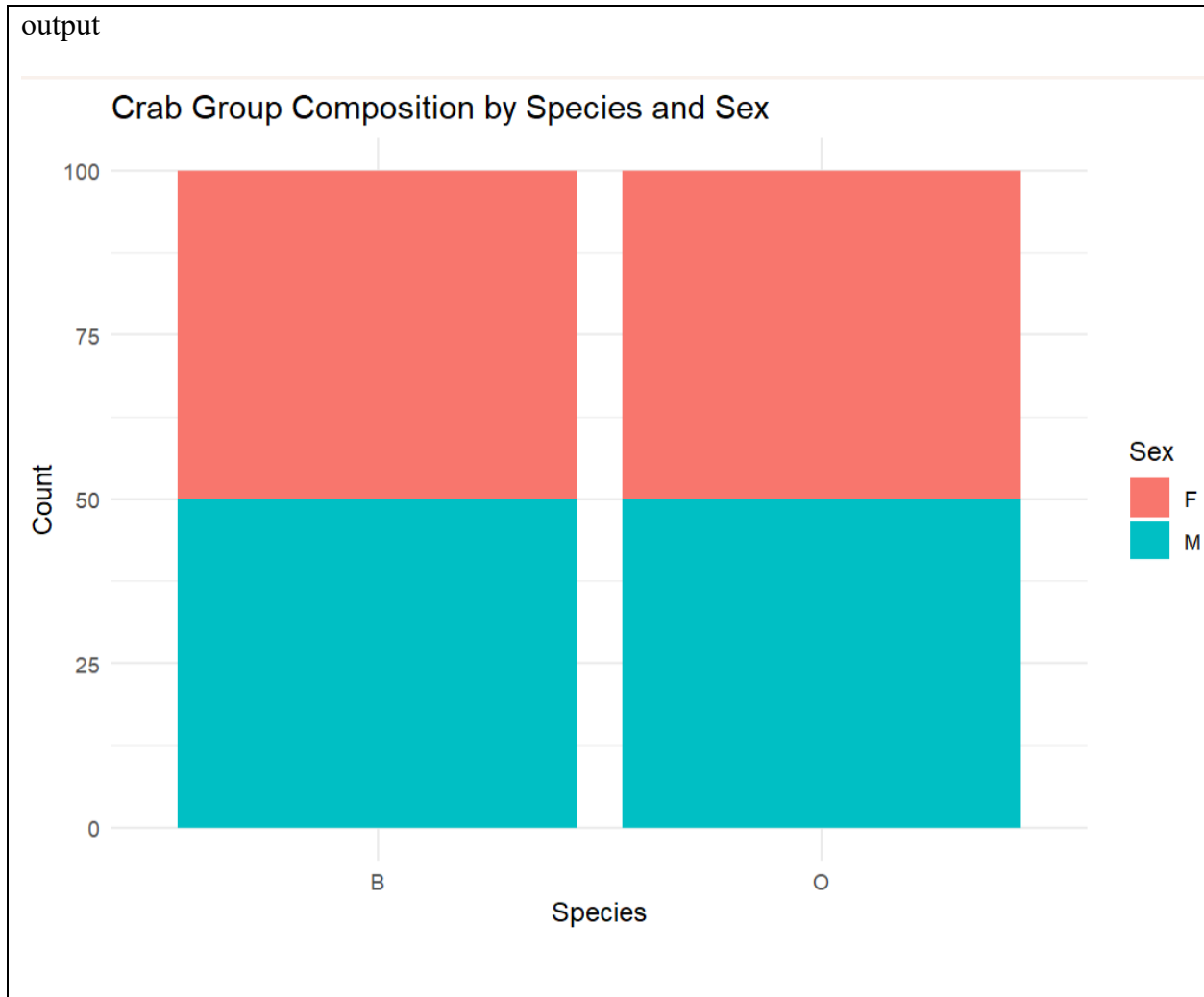
```
Code:
table(crabs$sp, crabs$sex)
```

```
output
> table(crabs$sp, crabs$sex)
```

```
  F M
B 50 50
O 50 50
```

d)

```
Code:
library(ggplot2)
ggplot(crabs, aes(x = sp, fill = sex)) +
  geom_bar(position = "stack") +
  labs(title = "Crab Group Composition by Species and Sex",
        x = "Species",
        y = "Count",
        fill = "Sex") +
  theme_minimal()
```



- e) The data is perfectly balanced across species and sex. Both sexes have 50 observation each and both species also have 50 observations each.

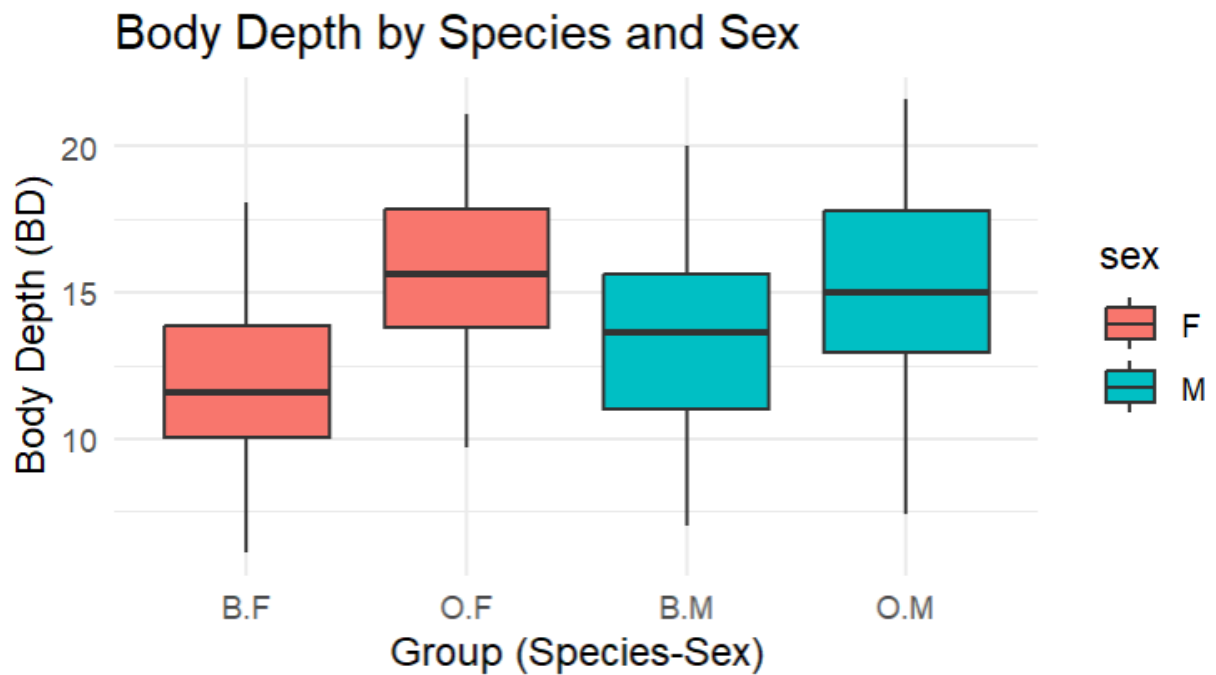
QUESTION 2

a)

Code

```
ggplot(crabs, aes(x = interaction(sp, sex), y = BD, fill = sex)) +  
  geom_boxplot() +  
  labs(title = "Body Depth by Species and Sex",  
        x = "Group (Species-Sex)",  
        y = "Body Depth (BD)") +  
  theme_minimal()
```

output



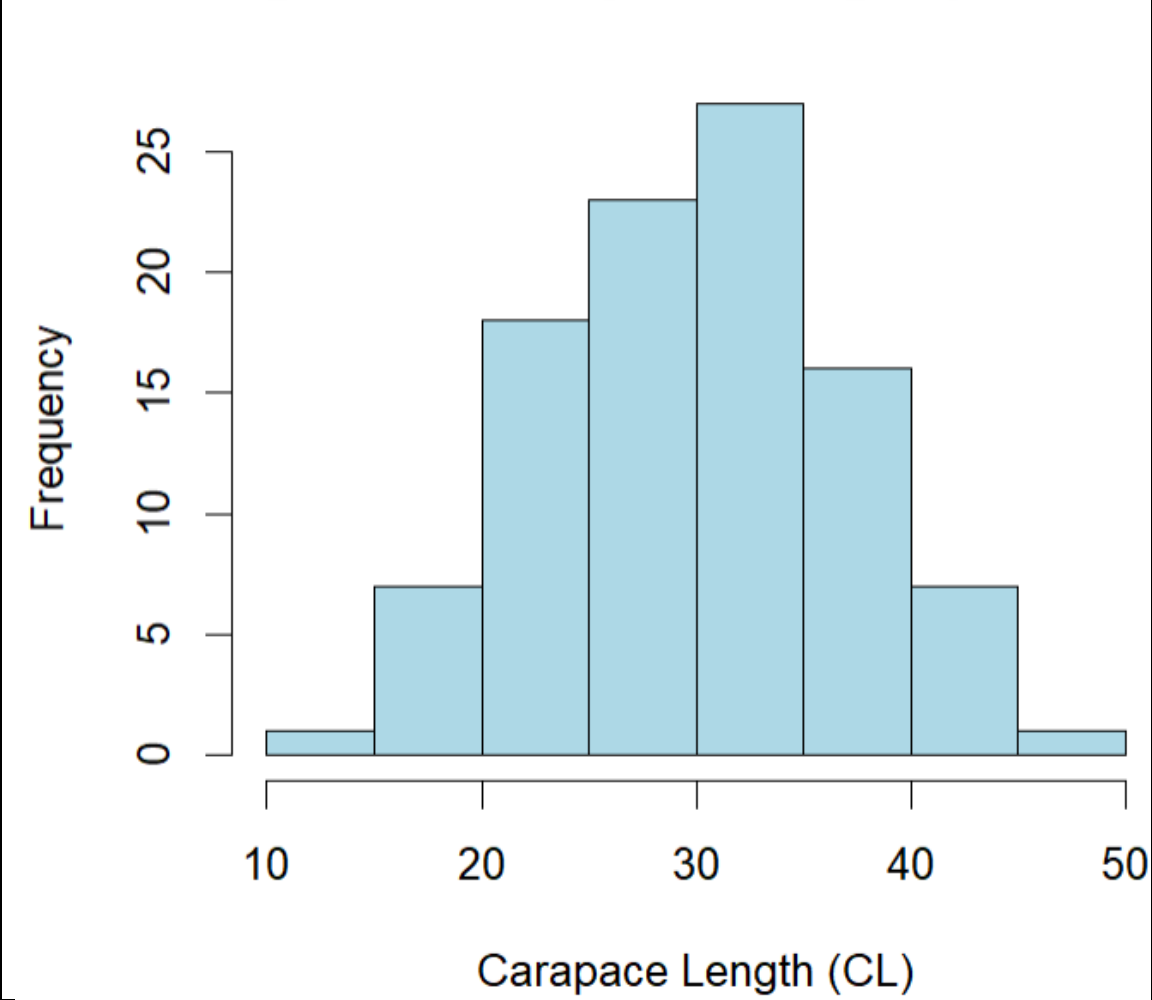
b) Kkkk

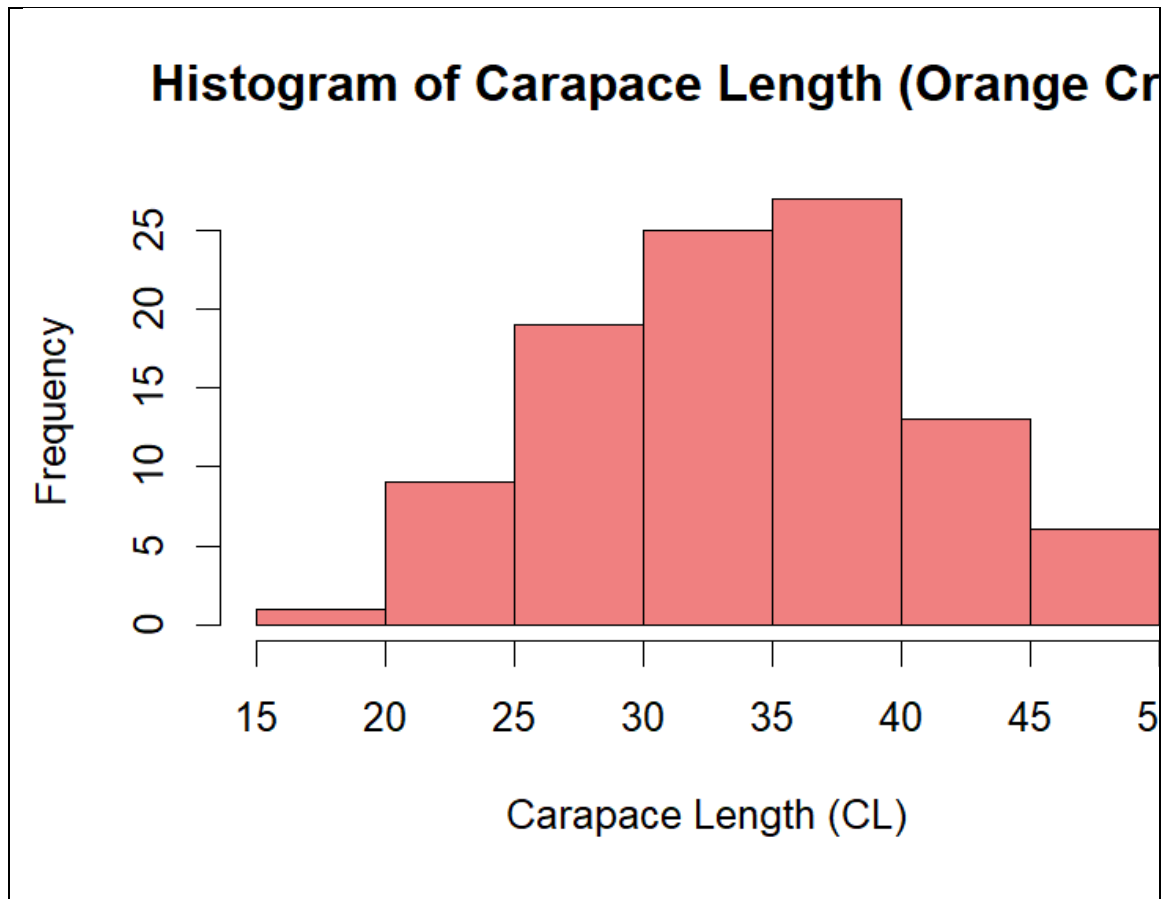
QUESTION 3

```
a) Codes
hist(crabs$CL[crabs$sp == "B"],
     main = "Histogram of Carapace Length (Blue
     Crabs)",
     xlab = "Carapace Length (CL)",
     col = "lightblue",
     border = "black")
hist(crabs$CL[crabs$sp == "O"],
     main = "Histogram of Carapace Length (Orange
     Crabs)",
     xlab = "Carapace Length (CL)",
     col = "lightcoral",
     border = "black")
```

output

Histogram of Carapace Length (Blue Cr)





- b) The histogram for both species show very similar shapes. Both shows single clear mode signifying that both are unimodal. Orange crabs are almost numeric while blue crab show a slight skewing to the right.

QUESTION 4

a) Codes

```
(means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
data = crabs, mean))
```

Output

```
> ##q4
> ## a) means of each morphological measurement by species and sex
> (means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex, data = crabs,
mean))
```

	sp	sex	FL	RW	CL	CW	BD
1	B	F	13.270	12.138	28.102	32.624	11.816
2	O	F	17.594	14.836	34.618	39.036	15.632
3	B	M	14.842	11.718	32.014	36.810	13.350
4	O	M	16.626	12.262	33.688	37.188	15.324

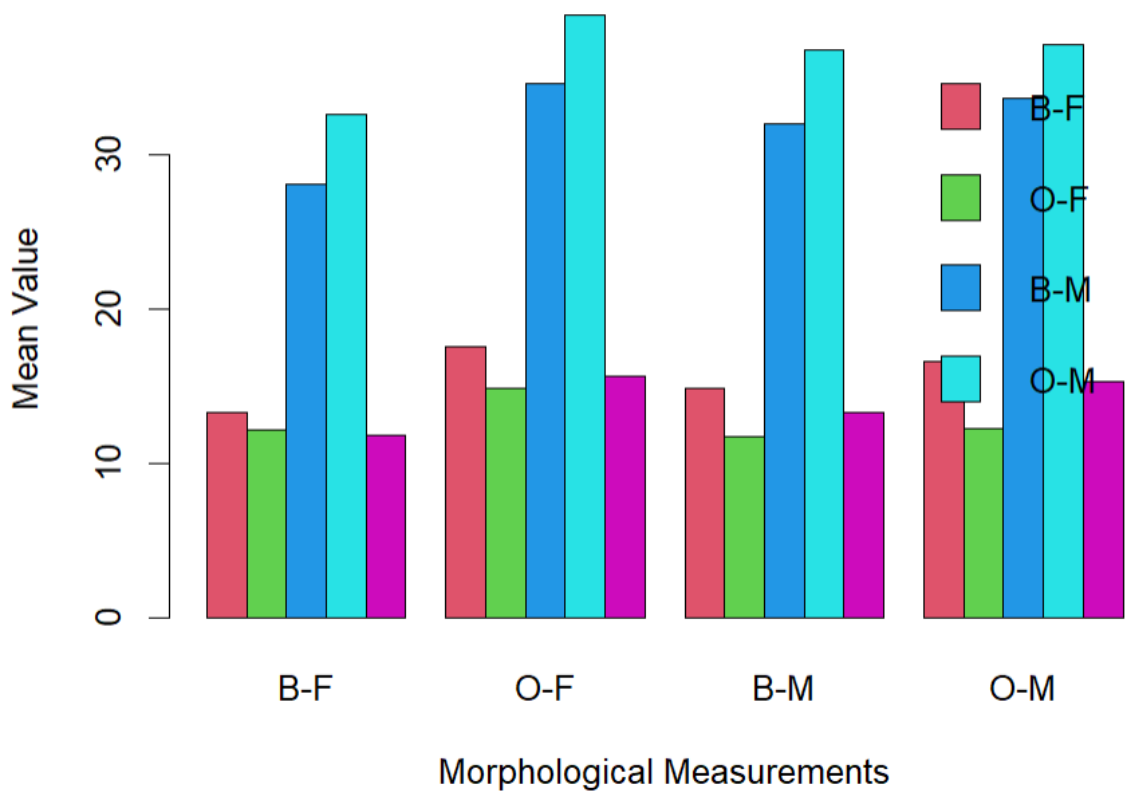
```

b) Codes
means_matrix<- as.matrix(means[, -c(1,2)])
rownames(means_matrix) <- paste(means$sp, means$sex,
sep = "-")
means_matrix
barplot(t(means_matrix),
        beside = TRUE,
        col = 2:6,
        legend.text = rownames(means_matrix),
        args.legend = list(x = "topright", bty =
"n"),
        main = "Mean Morphological Measurements by
Species and Sex",
        ylab = "Mean Value",
        xlab = "Morphological Measurements")

```

output

Mean Morphological Measurements by Species and Sex

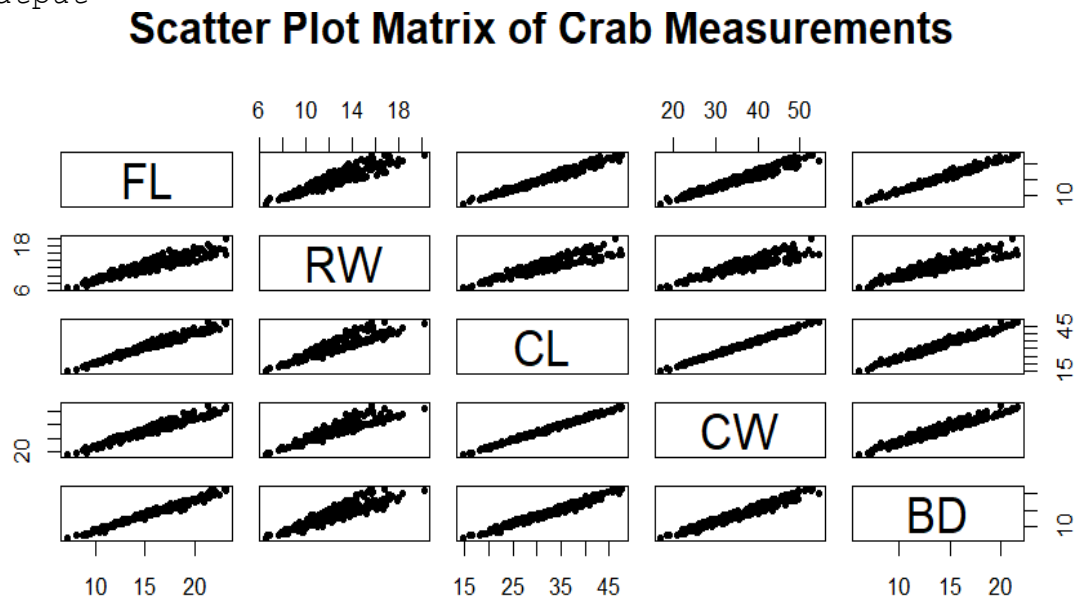


c) Hhhhh

QUESTION 5

```
a) Codes
crabs_measurements<-crabs[, c("FL", "RW", "CL", "CW",
"BD")]
pairs(crabs_measurements,
      main = "Scatter Plot Matrix of Crab Measurements",
      pch = 20, col = "black")
```

output



b) The variables show strong linear relationships.

QUESTION 5

The data has 200 observations categorized by sex and species. Each species has exactly 50 observations showing perfectly balanced data. Body depth as examined across the four species-sex groups using side-by-side boxplots shows distributions are unimodal and approximately symmetric.